AFFINE MORPHISMS: QC PUSHFORWARD IS EXACT, VANISHING, AND COMMUTES WITH COLIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Affine case vanishing. Stacks Project, Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-loc-qcoh-fbc-affine-direct-image. Gives \$R^i f_{QCoh,*} = 0\$, i > 0, for affine morphisms.
- Derived pushforward of qc modules is a delta-functor. Cohomology of Algebraic Stacks, proposition-direct-image-quasi-coherent (L30092 in this lane). Provides the delta-functor structure used to conclude exactness.
- Direct sums commute with qc pushforward. Stacks Project, Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-quasi-coherent-pushforward-direct-sums. Handles direct sums; with exactness this yields all small colimits.

Source reference. Original lemma 0GQI: sources/stacks-chow/tags.txt, source lines 20863–20863. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 077A: sources/stacks-chow/tags.txt, source lines 9195–9195. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0GQX: sources/stacks-chow/tags.txt, source lines 20877–20877. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be an affine morphism of algebraic stacks. The functors $R^i f_{QCoh,*}$, $i > 0$ vanish and the functor $f_{QCoh,*}$ is exact and commutes with direct sums and all colimits.*

Proof. Since we have $R^i f_{QCoh,*} = Q \circ R^i f_*$ we obtain the vanishing from Lemma 0GQI. The vanishing implies that $f_{QCoh,*}$ is exact as $\{R^i f_{QCoh,*}\}_{i \geq 0}$ form a δ-functor, see Proposition 077A. Then $f_{QCoh,*}$ commutes with direct sums for example by Lemma 0GQX. An exact functor which commutes with direct sums commutes with all colimits. □

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.